

TASK 3 REPORT

Car Price Prediction using Machine Learning

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Internship

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1. Introduction

Car price prediction is one of the most common real-world applications of Machine Learning. The objective of this project is to develop a regression model capable of predicting the selling price of a used car based on several vehicle-related attributes.

This project helps understand data preprocessing, feature engineering, regression algorithms, and model evaluation techniques.

2. Objective

The primary objectives of this project are:

- To analyze car sales data.
 - To preprocess and clean the dataset.
 - To create useful features for prediction.
 - To train a Machine Learning regression model.
 - To evaluate prediction accuracy.
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3. Dataset Description

The dataset contains information about used cars.

Features

- Car_Name
- Year

- Selling_Price
- Present_Price
- Driven_kms
- Fuel_Type
- Selling_type
- Transmission
- Owner

Total Records: 301

4. Tools and Technologies

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-Learn
 - Jupyter Notebook
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5. Methodology

Step 1: Data Loading

The dataset was loaded using Pandas and inspected for structure and data types.

Step 2: Data Cleaning

- Checked for missing values.
- Verified data consistency.

Step 3: Feature Engineering

A new feature called Car_Age was created using:

$\text{Car_Age} = \text{Current Year} - \text{Manufacturing Year}$

The Year column was removed after feature extraction.

Step 4: Data Encoding

Categorical variables such as Fuel_Type, Selling_type, and Transmission were converted into numerical format using one-hot encoding.

Step 5: Exploratory Data Analysis

Correlation analysis was performed using a heatmap to identify relationships among variables.

Step 6: Model Training

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

Linear Regression was used for model training.

Step 7: Prediction and Evaluation

Predictions were generated on test data and evaluated using:

- Mean Absolute Error (MAE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score
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6. Visualizations

The following visualizations were created:

- Correlation Heatmap
 - Selling Price Distribution
 - Actual vs Predicted Price Scatter Plot
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7. Results

The Linear Regression model successfully predicted car prices using the provided features.

The Actual vs Predicted plot demonstrated a strong positive relationship between actual and predicted values, indicating good model performance.

8. Applications

This system can be used by:

- Car dealerships
- Online vehicle marketplaces
- Buyers
- Sellers
- Automotive analytics companies

to estimate fair market prices for used vehicles.

9. Conclusion

This project successfully implemented a Machine Learning-based Car Price Prediction System using Linear Regression. Through preprocessing, feature engineering, model training, and evaluation, accurate predictions were achieved. The project demonstrates practical applications of predictive analytics in the automotive industry.

10. Future Scope

- Random Forest Regression
- XGBoost Regression
- Hyperparameter Optimization
- Web Application Deployment
- Real-Time Price Prediction System

Thank You